Abstract

RescueNet AI: A Multi-Agent Disaster Prediction, Emergency Coordination, and Resource Allocation Framework

RescueNet AI is a Streamlit-based intelligent disaster management application designed to assist authorities in disaster prediction, emergency coordination, and efficient resource allocation. The system is developed using Python and integrates Pandas and NumPy for data processing, Scikit-learn for machine learning–based disaster prediction, Plotly for interactive analytics, and Folium with Streamlit for geospatial visualization.

The application follows a multi-agent framework consisting of a Prediction Agent, Coordination Agent, Allocation Agent, and Monitoring Agent. The Prediction Agent analyzes historical and environmental data such as rainfall, temperature, flood indicators, and seismic information to estimate disaster probability and severity. The Coordination Agent manages incident reporting, rescue-team assignment, and status tracking during emergencies. The Allocation Agent prioritizes affected regions and distributes available resources including medical kits, food supplies, water tankers, shelters, and transport units based on urgency and availability. The Monitoring Agent continuously updates the operational status and displays affected areas on an interactive map.

The entire framework is implemented as a multi-page Streamlit dashboard that functions as a centralized command center. The dashboard provides live alert panels, risk indicators, Plotly-based charts, resource-monitoring cards, and Folium-based disaster maps, enabling users to visualize predictions, monitor rescue operations, and make informed decisions in real time.

By combining machine learning, data analytics, geospatial mapping, and an enhanced Streamlit user interface, RescueNet AI offers a scalable and user-friendly solution for academic and prototype-level disaster management systems. The proposed framework aims to improve situational awareness, reduce emergency response time, optimize the use of limited resources, and strengthen disaster preparedness and resilience.

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